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Operating System

Laboratory Session: Server Health Checkup

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Server Health Checkup

- ▶ Regularly monitor the server's health to ensure optimal performance.
- ▶ Use diagnostic tools to identify bottlenecks
- ▶ Check for hardware issues
- ▶ Monitor network latency.
- ▶ Ensure backup systems are functional.



Server Health Checkup

- ▶ Regularly monitor the server's health to ensure optimal performance.
- ▶ Use diagnostic tools to identify issues.
- ▶ Schedule regular maintenance checks.



CPU Utilization

- ▶ Monitor CPU utilization of critical processes like 'java', 'http', 'mysql', etc.
- ▶ Set up alerts for high CPU usage.
- ▶ Coordinate with the application team for fine-tuning.



Zombie Processes

- ▶ Check for processes where the PID still exists in the process table after termination.
- ▶ Zombie processes can degrade server performance.
- ▶ Use system utilities to kill zombie processes.



Load Average

- ▶ Monitor the load average to tune the server for performance.
- ▶ High load averages indicate performance issues.
- ▶ Consider load balancing solutions.



Disk/SAN/NAS Utilization

- ▶ Check the I/O reports for externally attached storage.
- ▶ Track the speed of read/write operations.
- ▶ Coordinate with storage and network teams for issues.

Monitoring CPU Utilization in Ubuntu

- ▶ Use the following shell script to monitor CPU utilization.

```
#!/bin/bash
# Monitor CPU utilization
while true; do
    top -b -n1 | grep 'Cpu(s)' | awk '{print $2 + $4}'
    sleep 5
done
```


Overview of Functions

- ▶ The `check_running_processes` Function
- ▶ **Example:** `ps aux --sort=-%cpu | awk 'NR<=5 {print $0}'`
 - ▶ Lists all running processes.
 - ▶ Sorts them in descending order based on CPU usage.
 - ▶ Uses `awk` to print the top 5 CPU-consuming processes.

Overview of Functions

- ▶ Using ps aux and awk
- ▶ Example: `ps aux | grep 'java'`
 - ▶ Lists all running processes.
 - ▶ Filters processes that contain the word 'java'.
 - ▶ Outputs the filtered list.

Overview of Functions

- ▶ The `check_cpu_utilization` Function
- ▶ **Example:** `ps -eo %cpu,command | egrep '(java|http|mysql)' | awk '$1 > 10 {print $0}'`
 - ▶ Lists CPU utilization and command name for each process.
 - ▶ Filters for 'java', 'http', and 'mysql' processes.
 - ▶ Prints processes that consume more than 10% CPU.

Overview of Functions

- ▶ Understanding `ps -eo %cpu,command`
- ▶ Example: `uptime | awk '{print $10}'`
 - ▶ Displays the system's uptime and load average.
 - ▶ Filters to show only the average CPU load in the last minute.
 - ▶ Outputs the filtered information.

Overview of Functions

- ▶ The `check_memory_utilization` Function
- ▶ **Example:** `sync && echo 3 > /proc/sys/vm/drop_caches`
 - ▶ Synchronizes cached data to disk.
 - ▶ Clears the system's cache.
 - ▶ Frees up memory resources.

Overview of Functions

- ▶ Using `free -h` and `swapon`
- ▶ Example: `free -h`
 - ▶ Displays memory usage.
 - ▶ Shows the information in human-readable format.
 - ▶ Outputs total, used, and free memory.

Overview of Functions

- ▶ The check_zombie_processes Function
- ▶ **Example:** `ps aux | awk '$8=="Z" {print "kill -9 " $2}' | sh`
 - ▶ Lists all running processes.
 - ▶ Filters for zombie processes.
 - ▶ Kills the identified zombie processes.

Overview of Functions

- ▶ **Zombie Processes and Their Impact**
- ▶ **Example:** `ps aux | awk '$8=="Z" {print $1}'`
 - ▶ Lists all running processes.
 - ▶ Filters for zombie processes.
 - ▶ Outputs the user owning the zombie processes.

Overview of Functions

- ▶ The `check_load_average` Function
- ▶ Example: `uptime`
 - ▶ Displays the system's uptime and load average.
 - ▶ Shows the load average for the last 1, 5, and 15 minutes.
 - ▶ Outputs the number of users and running processes.

Overview of Functions

- ▶ Understanding uptime
- ▶ Example: `uptime | awk '{print $11}'`
 - ▶ Displays the system's uptime and load average.
 - ▶ Filters to show only the load average for the last 5 minutes.
 - ▶ Outputs the filtered information.

Overview of Functions

- ▶ The `check_disk_utilization` Function
- ▶ Example: `iostat`
 - ▶ Displays disk I/O statistics.
 - ▶ Shows read/write speed and other metrics.
 - ▶ Outputs the information for each disk.

Overview of Functions

- ▶ Using iostat and df -h
- ▶ Example: `df -h`
 - ▶ Displays disk usage.
 - ▶ Shows the information in human-readable format.
 - ▶ Outputs total, used, and available disk space.

Summary

▶ Summary and Best Practices

- ▶ We summarize the key takeaways from the script and provide best practices for server health monitoring.
- ▶ **Note:** No specific shell script commands are used here.

Breaktime

